

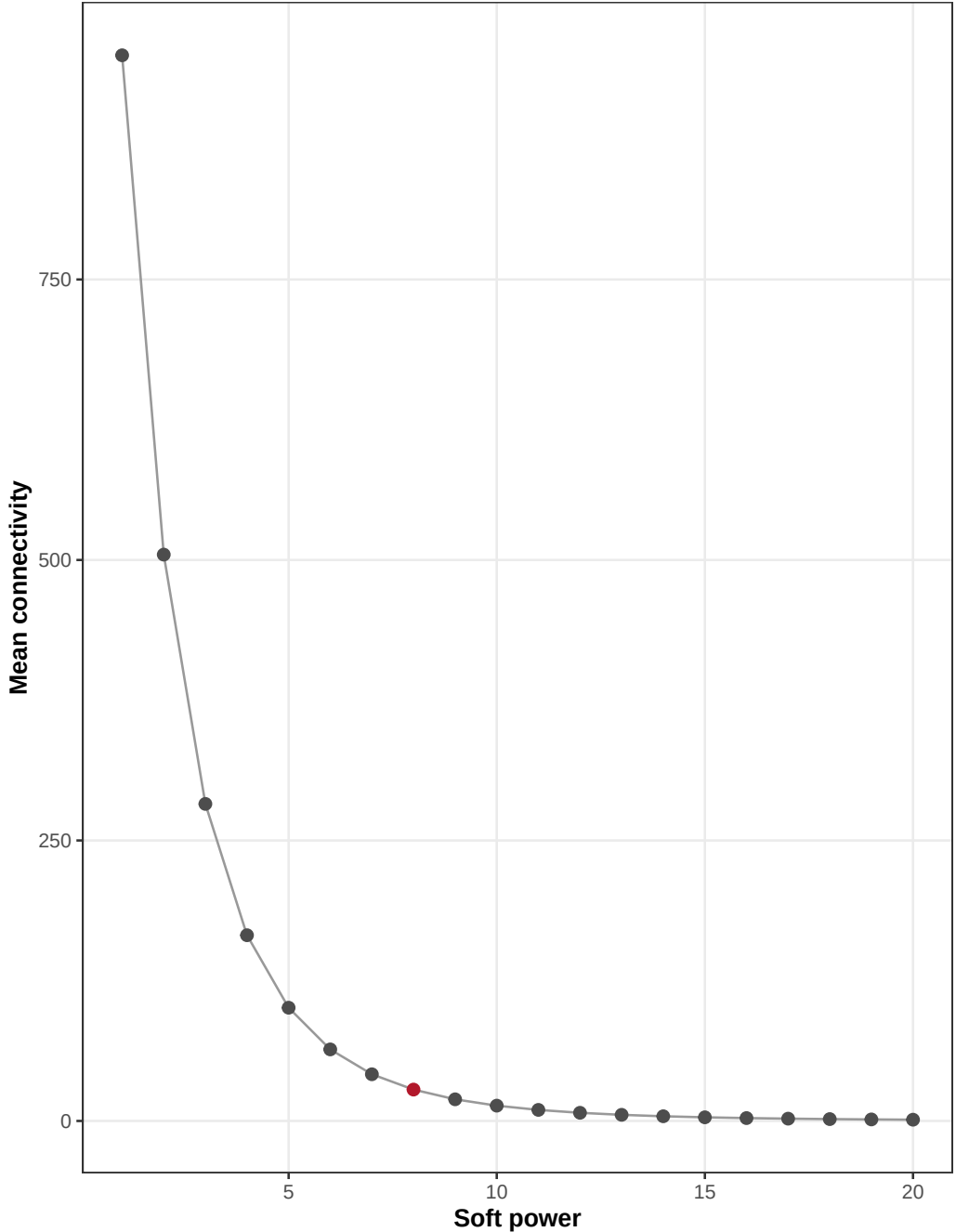
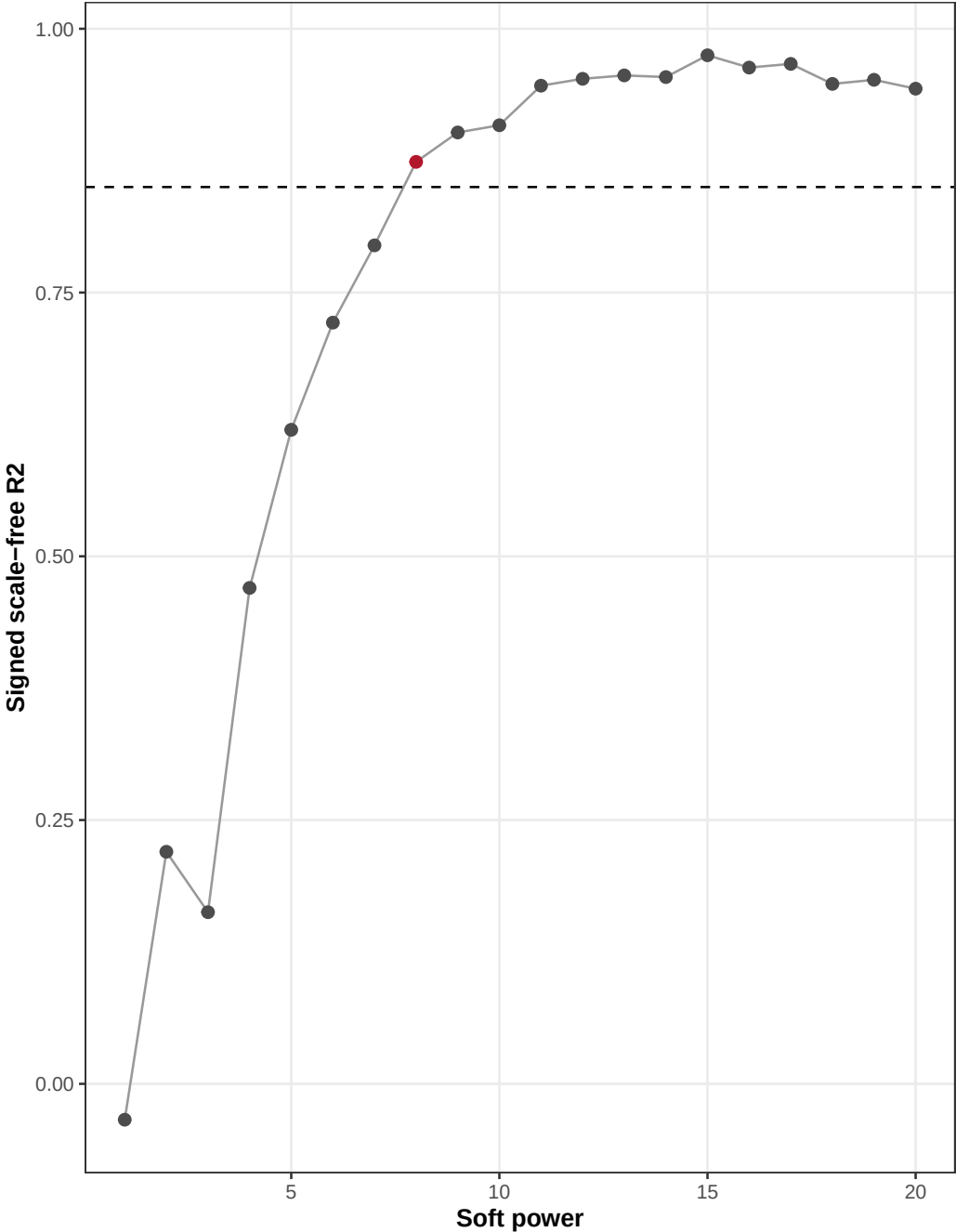
HRvLR 04 Networks

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Soft-threshold choice

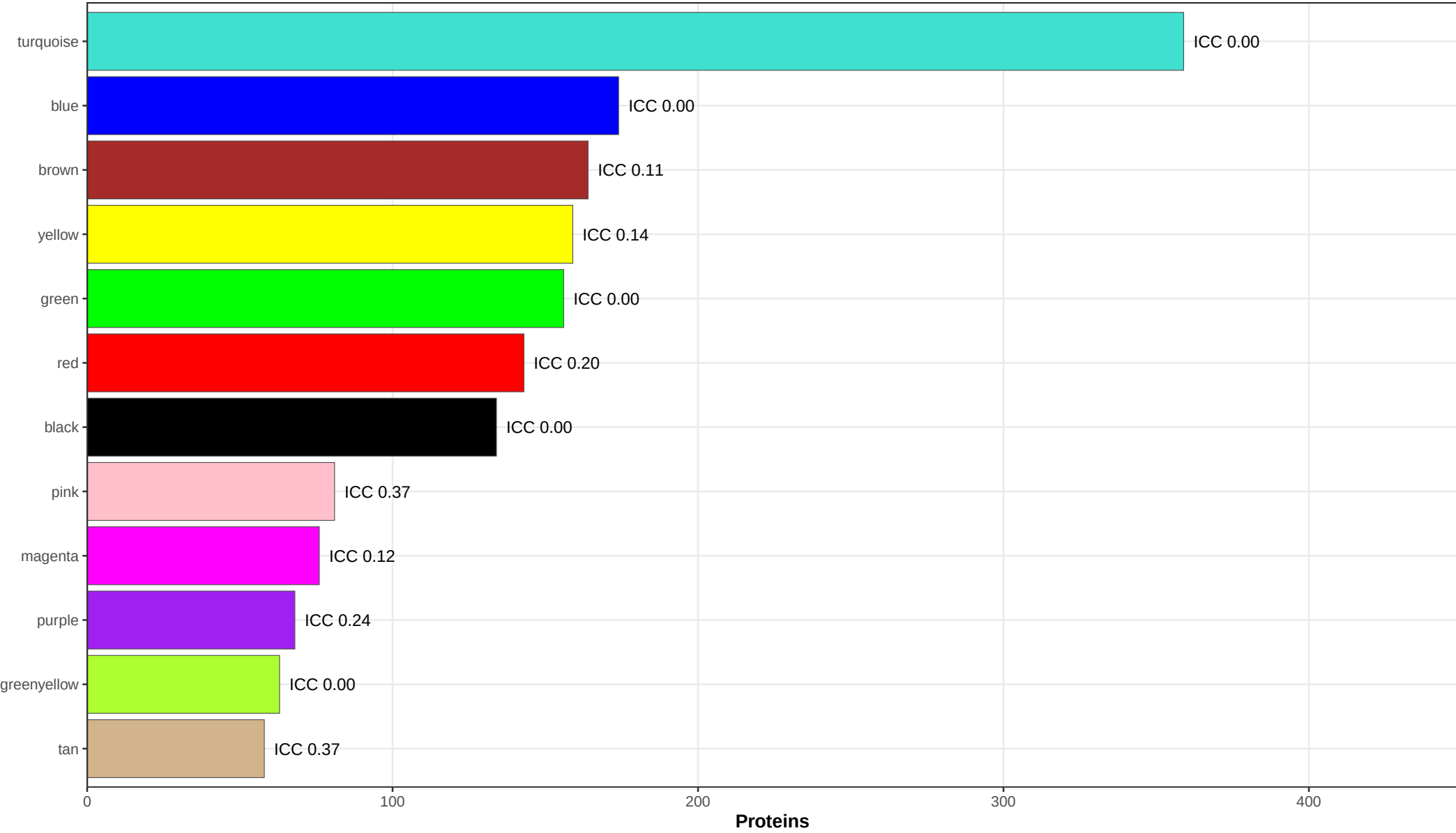
WGCNA signed, bicor, subject-centred; power 8 (R2 0.87, k 28.0)



Left: scale-free fit at each power; dashed line is pickSoftThreshold's 0.85 criterion, and the red point is the lowest power that clears it. Right: mean connectivity at each power. Data: 04_Networks/01_Modules/c_data/modules.xlsx, sheet soft_threshold.

Module sizes and subject dependence

12 modules; 265 of 1900 proteins unassigned (grey)



Bars: proteins per module. Label: intraclass correlation of the eigengene across a subject's three biopsies (lme4, 1 | subject). Modules are defined on within-subject-centred data so that subject identity cannot build them; a high ICC here means the scored eigengene still differs between people.

Hubs (highest kME): black: PSMA7, CAPZA1, PTGES3, DARS1, RAB14, RTN2, CRKL, SRM, CZIB, FKBP2; blue: ETFDH, DIABLO, SUCLA2, SIRT5, MT-ATP8, ACAT1, COQ7, RPS6KA3, COA6, IDI1; brown: PFN1, AHNAK, CANX, HSPA5, PPIA, CAP1, ARHGDIA, IFITM1, HPRT1, ACTR3; green: RARS1, MYLK, BZW2, CAB39L, ME1, NIF3L1, AARSD1, ENDOD1, GCSH, PRXL2A; greenyellow: AOC3, LUM, CAV2, DDAH2, HPCAL1, TAGLN, CD9, SEPTIN11, TMED10, UGGT1; magenta: RPL7, RPLP2, RPLP0, RPL7A, RPL18, RPL12, RPL10A, RPL18A, LDB3, RPS16; pink: TTN, NEB, TPM1, TNNC2, MYH7, MYBPC1, TPM3, MYL3, MYH2, ACTN2; purple: RPL23A, RPSA, RPL5, RPS18, VAPA, RPS6, RPS4X, RPL19, RPS12, RPS9; red: YWHAG, REEP5, EEF2, PHPT1, HEBP2, EEF1A1, ARL8B, CA3, PRKCSH, YWHAH; tan: NDUFS1, NDUFV1, NDUFS8, NDUFB7, NDUFB3, NDUFS3, IMMT, CHCHD3, NDUFV2, NDUFA13; turquoise: GOSR2, DRG1, MRPL9, TIPRL, GCN1, SMIM11A, SPR, PYURF, PXN, PPP2R1A; yellow: SAMM50, DLAT, PDHA1, TFAM, TMEM11, MT-ND3, TMEM143, NDUFB9, RPS13, SLC25A6. Data: 04_Networks/01_Modules/c_data/modules.xlsx, sheet subject_icc.

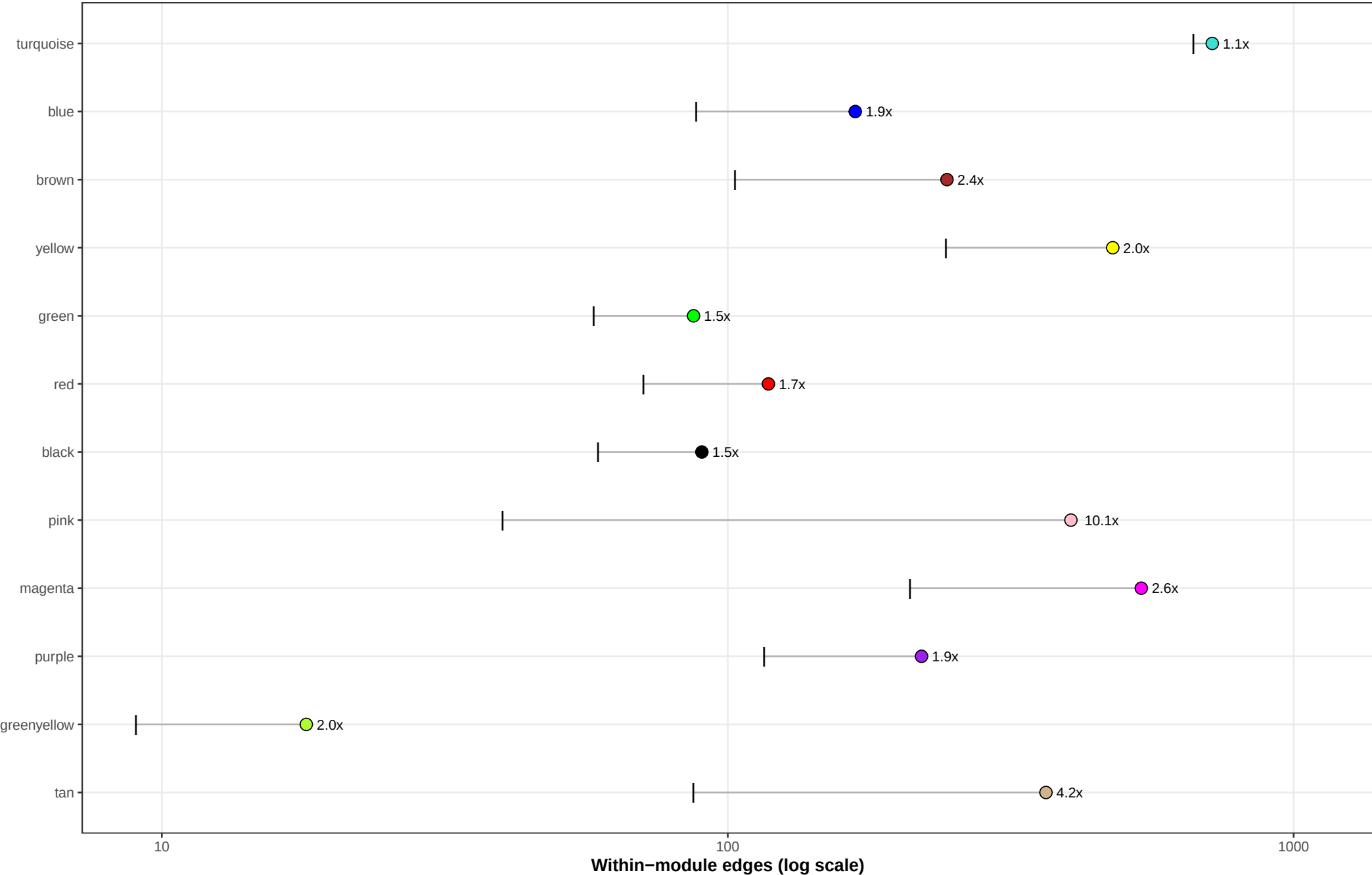
What each module is: enrichment

clusterProfiler ORA, universe 1900; top 3 sets per module, BH < 0.05



enrichplot::dotplot of the compareCluster result. Columns: modules (members with a set in brackets). Dot: one of the three sets with the smallest p in that module at BH < 0.05; size = gene ratio, colour = BH within module; modules with none are absent. Hubs per module: caption of the module-size page. Data: 04_Networks/02_Characterise/c_data/module_characterisation.xlsx, sheet ora.

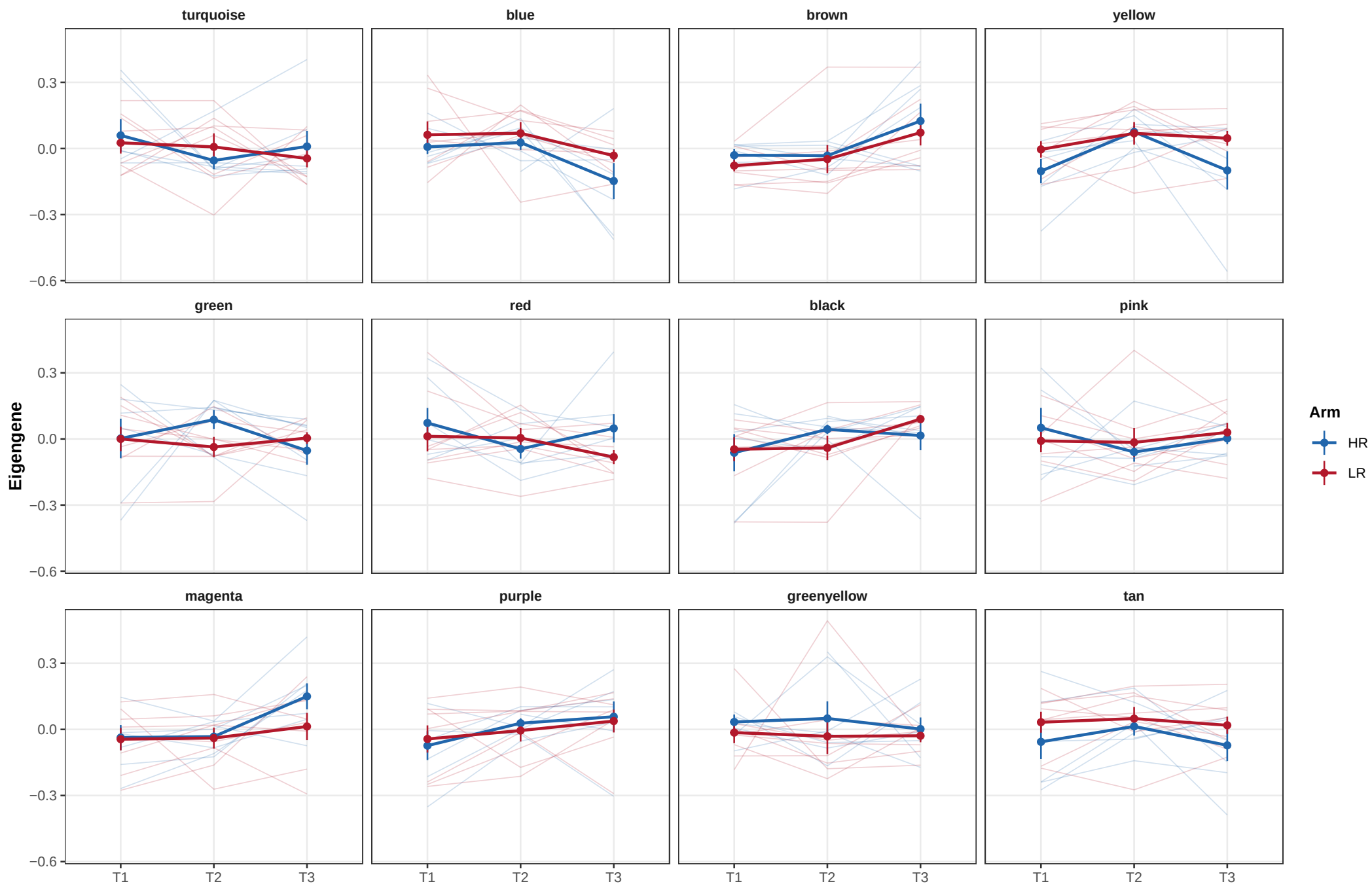
STRING edges within each module
STRINGdb PPI enrichment, v12, score >= 700; detected background



Point: observed high-confidence STRING edges among a module's members. Tick: edges STRINGdb expects from the members' degrees in the detected background. Label: observed / expected. Modules at BH < 0.05: 12 of 12. Data: 04_Networks/02_Characterise/c_data/module_characterisation.xlsx, sheet string.

Eigengene trajectories

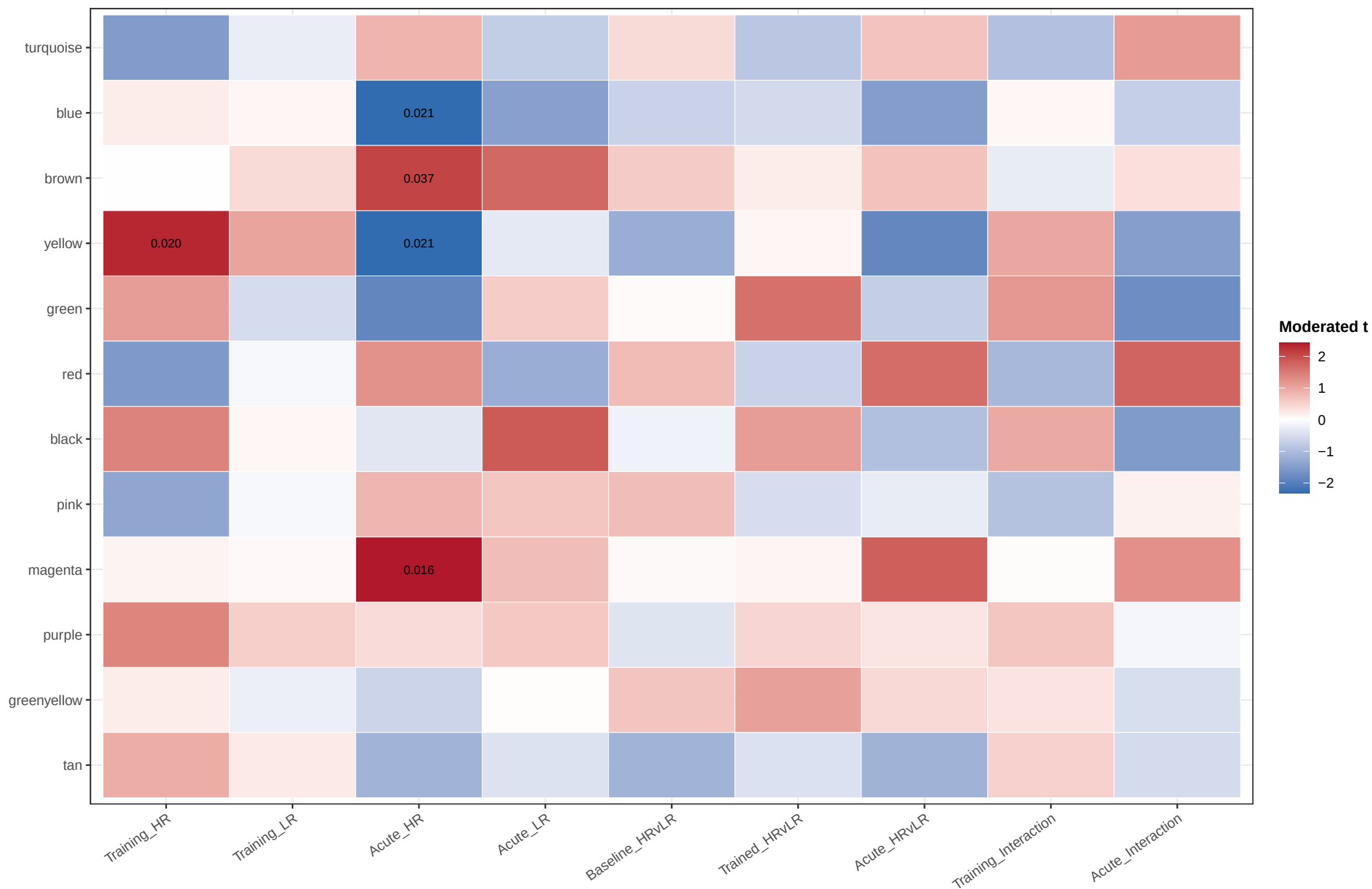
Module eigengene per biopsy, scored on raw abundance



Thin lines: one subject's three biopsies. Thick line and range: arm mean $\pm$ standard error at each timepoint. Data: 04_Networks/01_Modules/c_data/modules.xlsx, sheet eigengenes.

Module eigengenes across the nine contrasts

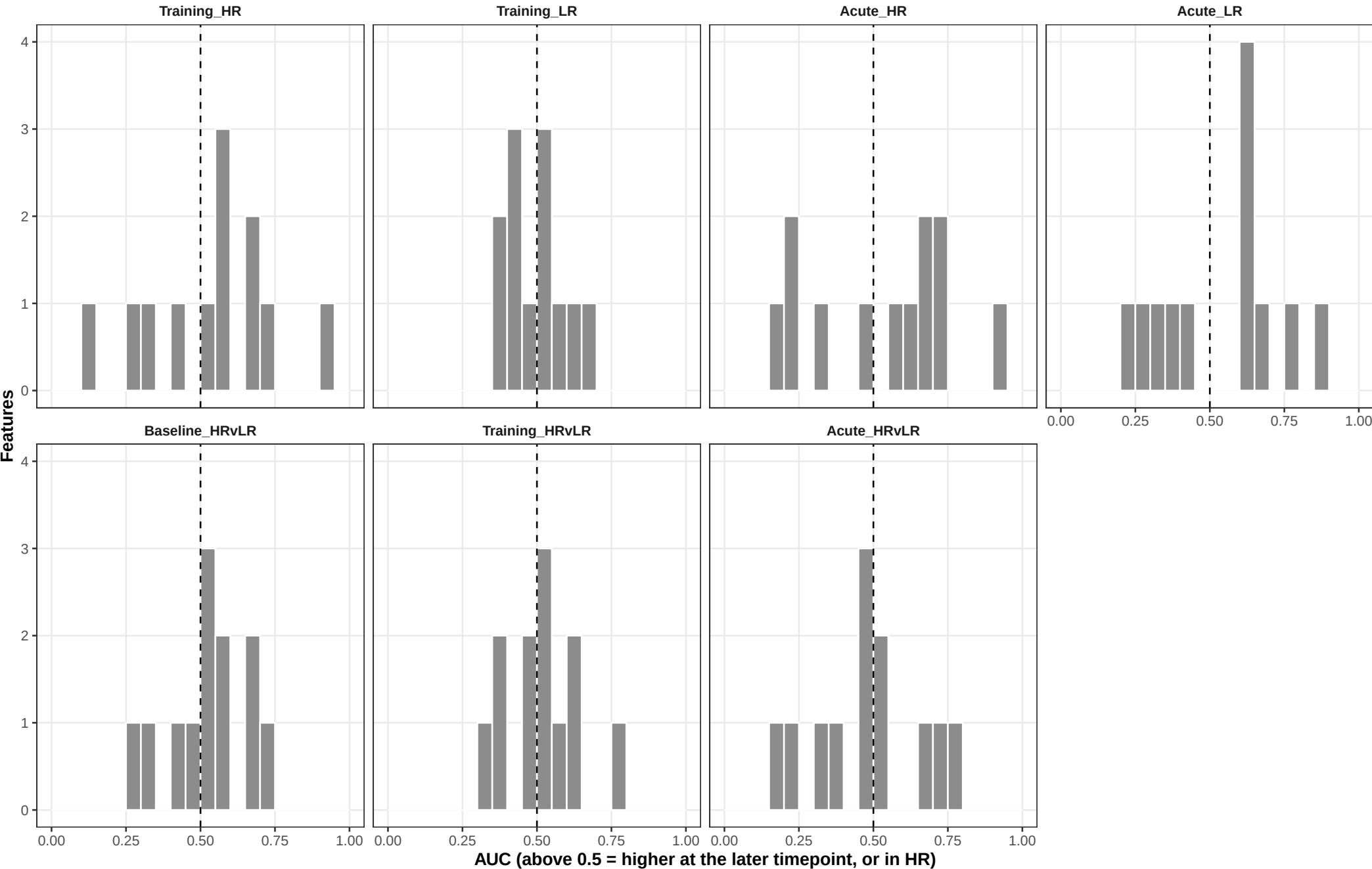
limma, protein design, subject block (cor 0.114); BH per contrast



Fill: moderated t of each module in each contrast. Label: nominal p where below 0.05. Eigengenes at BH < 0.05 across all contrasts: 0. Data: 04_Networks/03_Classify_Associate/c_data/module_screens.xlsx, sheet contrasts.

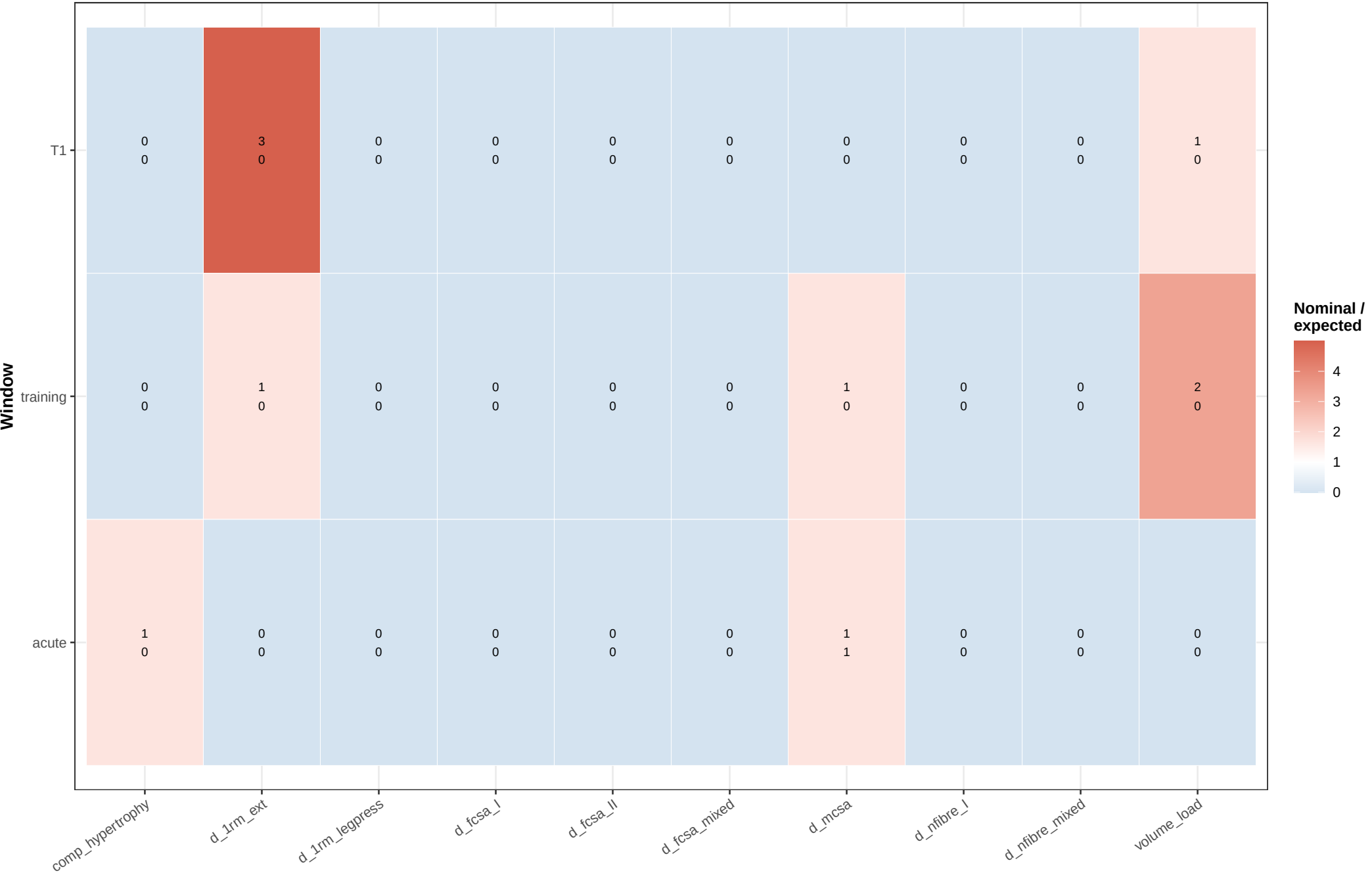
Module AUC per classification task

Mann-Whitney AUC, Wilcoxon p (signed-rank within arm); 12 features



Each feature contributes one AUC per task. Within-arm tasks compare a subject's later timepoint with its earlier one; HRvLR tasks compare subjects on the baseline level or on their own change. Dashed line: chance. Data: 04_Networks/03_Classify_Associate/c_data/module_screens.xlsx, sheets classify and chance_classify.

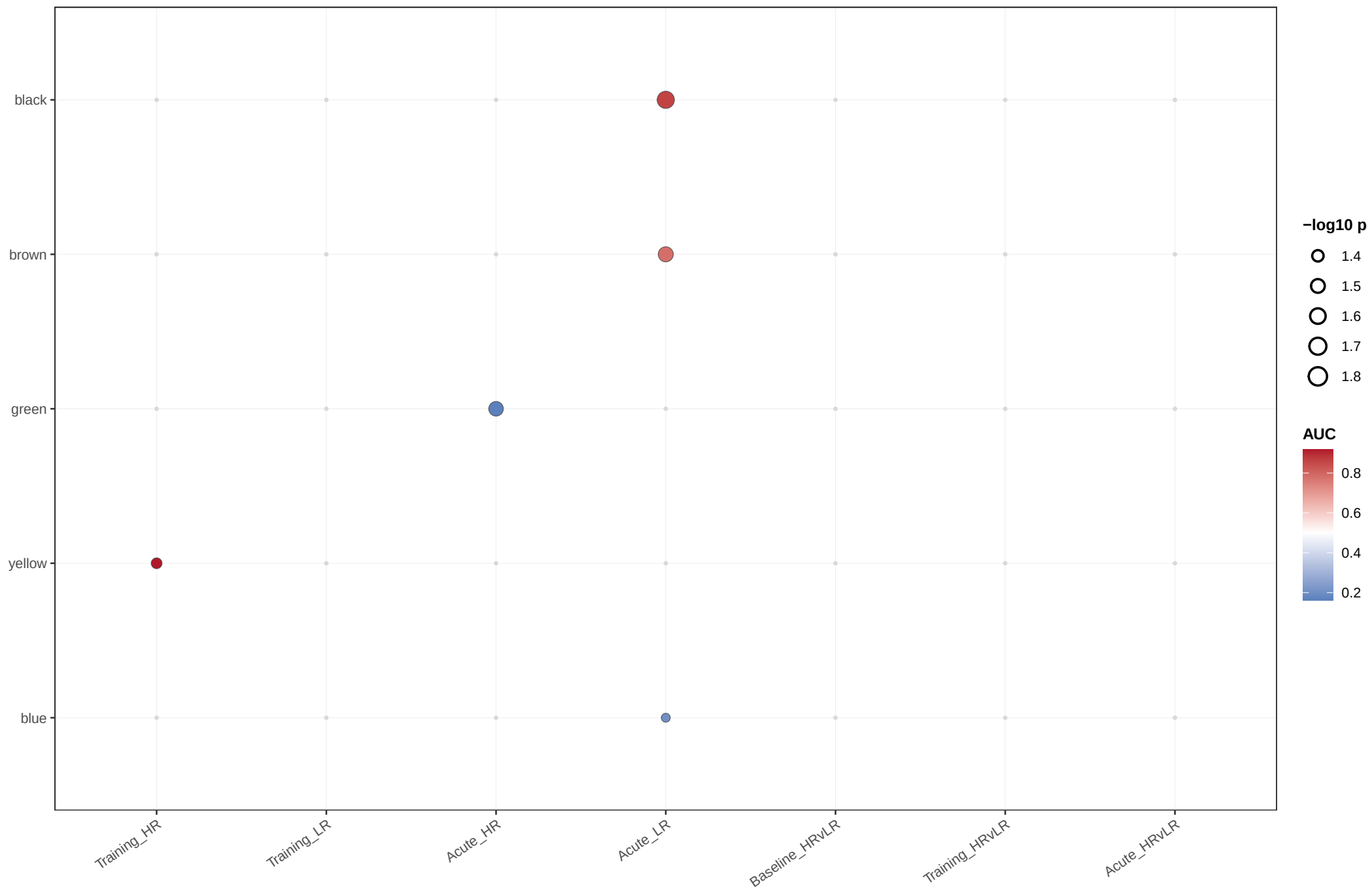
Module association with phenotype
limma on a continuous predictor; BH per cell; 12 features



Rows: the feature at T1, or its change over training (T2 – T1) or the acute bout (T3 – T2). Columns: the ten phenotypes. Fill: nominal hits at $p < 0.05$ over the count expected by chance ($0.05 \times$ tested); white is what an empty screen looks like. Top number: nominal hits; bottom: BH < 0.05 . Data: 04_Networks/03_Classify_Associate/c_data/module_screens.xlsx, sheets associate and chance_associate.

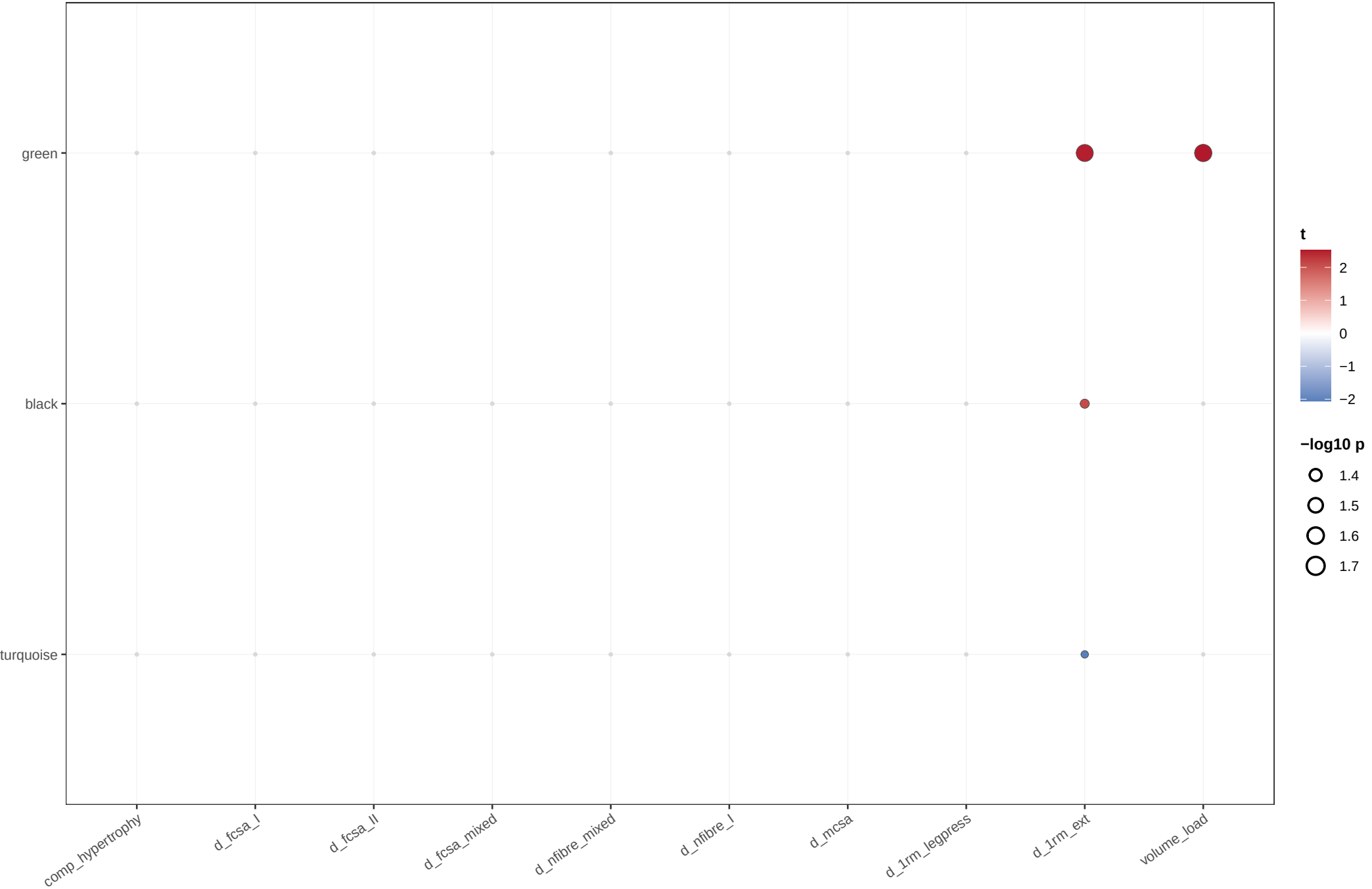
Module classification hits (1/1)

AUC and Wilcoxon p per task, paired within arm; 5 features nominal in at least one column; rows 1–5



Module association hits, level at T1 (1/1)

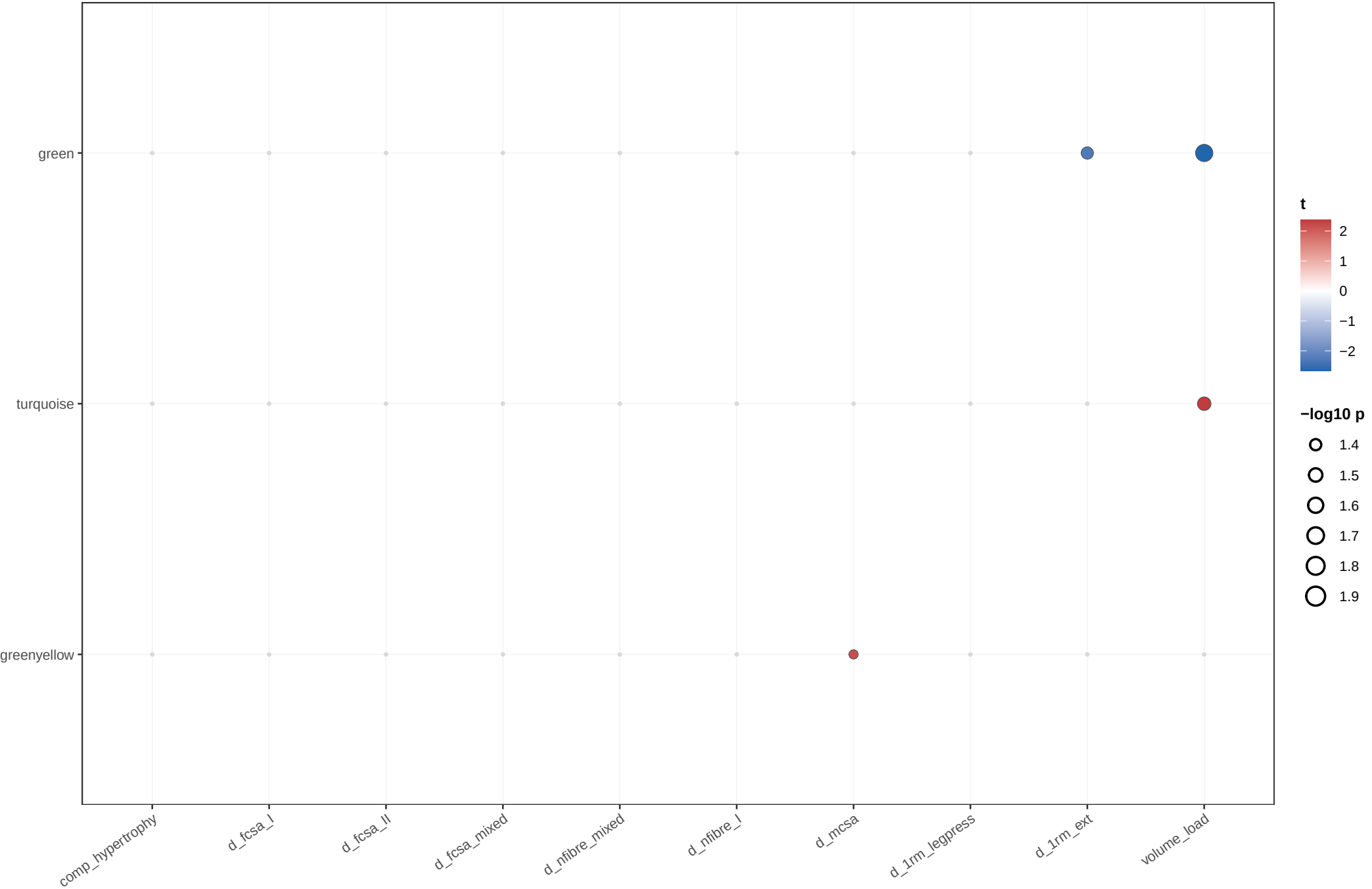
limma slope t against each phenotype, one row per subject; 3 features nominal in at least one column; rows 1–3



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t , size = $-\log_{10} p$. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 04_Networks/03_Classify_Associate/c_data/module_screens.xlsx, sheet associate.

Module association hits, training change (T2 – T1) (1/1)

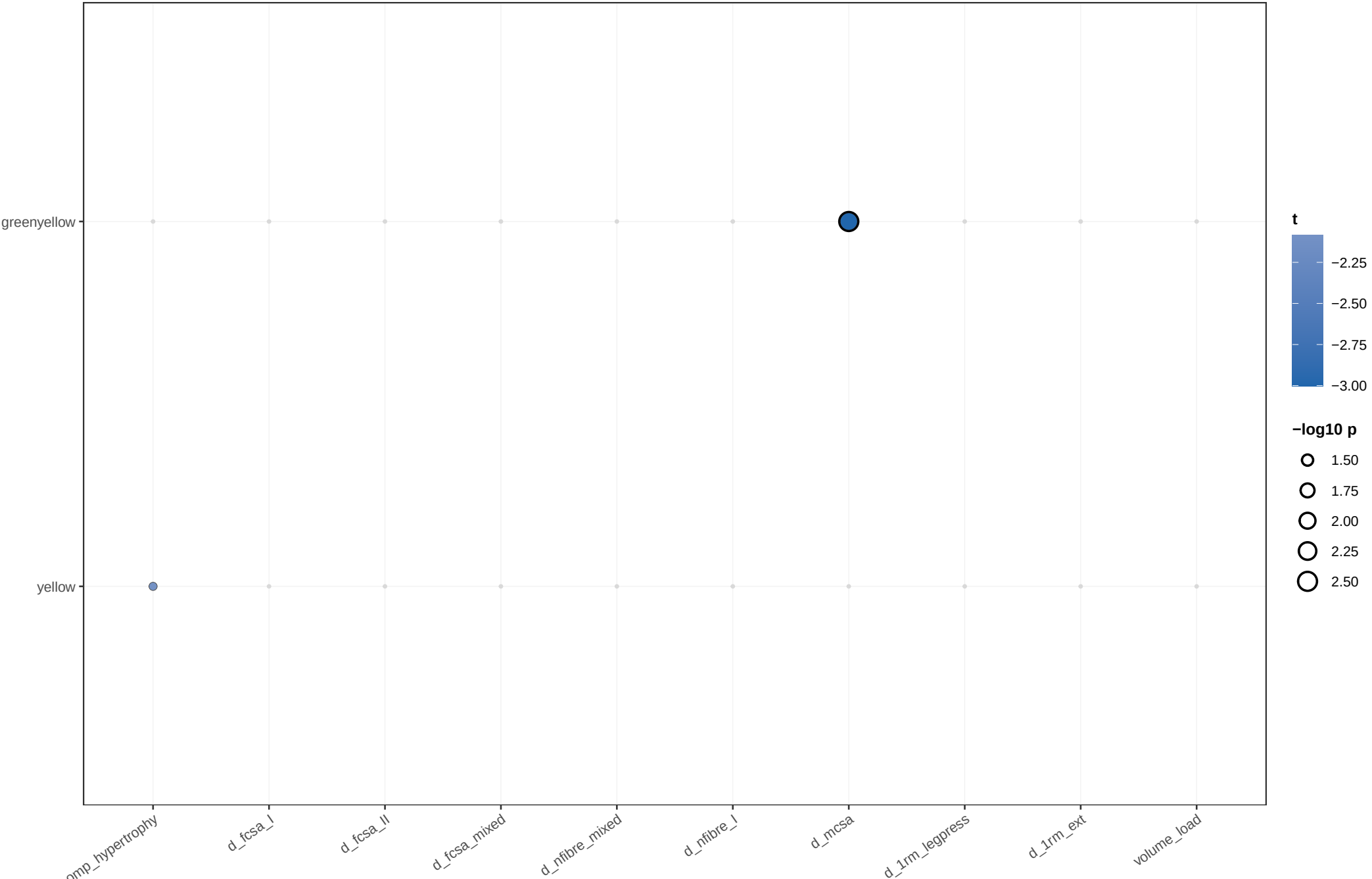
limma slope t against each phenotype, one row per subject; 3 features nominal in at least one column; rows 1–3



Rows: features at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal in that column, fill = t, size = -log10 p. Black ring: BH < 0.05. Grey speck: tested, not nominal. Data: 04_Networks/03_Classify_Associate/c_data/module_screens.xlsx, sheet associate.

Module association hits, acute change (T3 – T2) (1/1)

limma slope t against each phenotype, one row per subject; 2 features nominal in at least one column; rows 1-2



Rows: features at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal in that column, fill = t, size = $-\log_{10} p$. Black ring: BH < 0.05 . Grey speck: tested, not nominal. Data: 04_Networks/03_Classify_Associate/c_data/module_screens.xlsx, sheet associate.